

ASSIGNMENT 3

Trees

Learning outcome:

1. Deals with Binary Search Tree (BST)
2. Implementation of the Binary Tree ADT using a linked tree structure

Dateline: 10 January 2026, upload to PutraBLAST

****** Copy or other forms of cheating are forbidden. The standard penalty for the first offense is to award 0 to all parties concerned.

Create a program that lets a user enter data, and builds a BST from that data. Your program will also let the user interact with that BST, using a series of commands:

- s number – this will search for “number” in the BST. Tell the user whether that number is found or not.
- d number – search for “number” and, if found, delete it. If it is not found, tell the user.
- X – this command is used to display information about the BST.
The following information should be displayed:
 - pre-, in- and post-order traversal of the tree. For each one, output the string “NLR:” “LNR:” or “LRN:” followed by a comma-separated list of the nodes. Output one line for each traversal.
 - output the height of the tree (an empty tree has height -1; a tree with a single node has height 0; etc.) Your output should say “HEIGHT:” followed by the height.
 - Indicate whether or not the tree is balanced. Your output should say “BALANCED:” followed by “YES” or “NO”
- ? – display a help message giving the user a list of valid commands.
- Q – exit the program

The number enter is a positive value, insert it into the tree, else not valid. Only allow a number to be entered once. If the user tries to insert a number that’s already in the tree, you should print an error message and not insert the second copy of the number. Typing anything else should print an error message and tell the user how to get help.